

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Summative Assessment 1

Summative Assessment (SA-1) - Answer Key

Class: Class 8

Assessment Type: Summative Assessment (SA-1)

Total Marks: 40

Duration: 2 hours

Topics Covered: Functions, Data Structures, SQL, HTML/CSS

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: MCQ (10 x 1 = 10 marks)

Q.No	Answer	Explanation
1	b	def keyword defines a function in Python
2	b	len([1, [2, 3], 4]) returns 3 (three top-level elements)
3	c	TRUNCATE deletes all rows but keeps table structure
4	a	<tr> tag creates a table row in HTML
5	c	ID selector #main has highest specificity
6	b	Output: [1] then [1, 2] due to mutable default argument
7	b	sorted() returns a new sorted list without modifying original
8	c	SELECT DISTINCT * FROM table; displays unique records
9	b	padding adds space between element border and content
10	c	List is mutable; tuple, string, integer are immutable

Part B: Short Answer (8 x 2 = 16 marks)

1. Write a Python function `reverse_string(s)`.

• **Answer:**

```
def reverse_string(s):
    """Return the reversed version of string s."""
    if not s:
        return ""
    return s[::-1]

# Test cases
print(reverse_string("hello"))    # Output: "olleh"
print(reverse_string(""))        # Output: ""
print(reverse_string("Python"))  # Output: "nohtyP"
```

• **Marking:** 1 mark for function definition, 1 mark for empty string handling.

2. Write the output and explain line by line.

• **Answer:** Output: 1, 120, 6. Line 1: `factorial(0)` returns 1 (base case). Line 2: `factorial(5)` = $5 \times 4 \times 3 \times 2 \times 1 = 120$. Line 3: `factorial(3)` = $3 \times 2 \times 1 = 6$.

• **Marking:** 1 mark for correct output, 1 mark for line-by-line explanation.

3. Differentiate between `list.append()` and `list.extend()`.

• **Answer:** `append()` adds a single element to the end (e.g., `list1.append([4,5])` adds `[4,5]` as one element). `extend()` adds each element individually (e.g., `list1.extend([4,5])` adds 4 and 5 separately).

• **Marking:** 1 mark for explaining `append`, 1 mark for explaining `extend`.

4. Write a Python program to count word frequency.

- **Answer:**

```
sentence = "the cat sat on the mat the cat"
words = sentence.split()
freq = {}
for word in words:
    freq[word] = freq.get(word, 0) + 1

for word, count in sorted(freq.items(), key=lambda x: x[1], reverse=True):
    print(f"'{word}': {count}")
```

- **Marking:** 1 mark for dictionary creation, 1 mark for frequency counting and display.

5. Write SQL queries for Employees table.

- **Answer:**

```
-- Find employees with salary between 30000 and 50000
SELECT * FROM Employees WHERE salary BETWEEN 30000 AND 50000;

-- Count employees per department
SELECT department, COUNT(*) FROM Employees GROUP BY department;

-- Find employee with maximum salary
SELECT * FROM Employees WHERE salary = (SELECT MAX(salary) FROM Employees);

-- Delete employees who joined before 2020
DELETE FROM Employees WHERE joining_date < '2020-01-01';
```

- **Marking:** 1 mark for each correct query (4 queries).

6. Differentiate between DELETE and TRUNCATE.

- **Answer:** DELETE: Removes specific rows, can use WHERE clause, can be rolled back, logs each row deletion. TRUNCATE: Removes all rows, faster than DELETE, cannot use WHERE clause, cannot be rolled back.
- **Marking:** 1 mark for three points about DELETE, 1 mark for three points about TRUNCATE.

7. Write HTML code for a form.

- **Answer:**

```
<form>
  <label for="name">Name:</label>
  <input type="text" id="name" name="name" required><br><br>

  <label for="email">Email:</label>
  <input type="email" id="email" name="email" required><br><br>

  <label for="dob">Date of Birth:</label>
  <input type="date" id="dob" name="dob" required><br><br>

  <label>Gender:</label>
  <input type="radio" id="male" name="gender" value="male">
  <label for="male">Male</label>
  <input type="radio" id="female" name="gender" value="female">
  <label for="female">Female</label><br><br>

  <input type="submit" value="Submit">
</form>
```

- **Marking:** 1 mark for form structure, 1 mark for all input types.

8. Write CSS rules for navigation, links, and media query.

- **Answer:**

```
/* Responsive navigation bar with flexbox */
nav {
  display: flex;
  justify-content: space-around;
  background-color: #333;
  padding: 10px;
}

nav a {
  color: white;
  text-decoration: none;
  padding: 10px 20px;
}

/* Hover underline effect */
nav a:hover {
  text-decoration: underline;
}

/* Media query for screens below 768px */
@media (max-width: 768px) {
  nav {
    flex-direction: column;
    align-items: center;
  }

  nav a {
    padding: 5px;
  }
}
```

- **Marking:** 1 mark for flexbox navigation, 1 mark for media query.

Part C: Long Answer (4 x 3 = 12 marks)

1. Write a complete Python program for a shopping cart.

- **Answer:** Complete program with functions for adding items, removing items, calculating total with GST, and displaying formatted receipt. Should include proper error handling and formatted output.
- **Marking:** 1 mark for add/remove functions, 1 mark for GST calculation, 1 mark for formatted receipt.

2. Write SQL queries for School database.

- **Answer:** Complex queries using JOINS, subqueries, and aggregate functions to find students scoring above 80, calculate percentages, find highest average subject, and rank students.
- **Marking:** 1 mark for each correct query approach (4 queries).

3. Create HTML page for Student Report Card.

- **Answer:** Complete HTML with header, styled table, summary statistics, CSS for professional formatting, and print-friendly layout.
- **Marking:** 1 mark for HTML structure, 1 mark for table styling, 1 mark for print-friendly CSS.

4. Write a Python Quiz Application.

- **Answer:** Program with dictionary for questions, input validation, score tracking, and results display with percentage and grade.

- **Marking:** 1 mark for question storage, 1 mark for input handling, 1 mark for score calculation.

Part D: Application Based (2 x 1 = 2 marks)

1. Identify the bug in the add_item function.

- **Answer:** Bug: Using mutable default argument cart=[]. The list persists between function calls. Corrected version: Use None as default and create new list inside function.
- **Marking:** 1 mark for identifying bug, 1 mark for corrected version.

2. Explain the SQL query step by step.

- **Answer:** Query selects department, counts employees, calculates average salary for employees who joined after 2020, groups by department, filters groups with average salary > 40000, sorts by average salary descending.
- **Marking:** 1 mark for explaining each clause's role.